

# The Immunity Factor from Contact Surveys: Where Epidemics Turn Over

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September 2026

## Abstract

Many COVID-19 waves reached their peak of incidence when estimated cumulative infections were well below the threshold  $1 - 1/R$  implied by their initial growth under a homogeneous model. Because people with many contacts are infected sooner, the effective reproduction number falls faster than the susceptible share  $S$ , approximately as  $RS^\lambda$ , where the immunity factor  $\lambda$  is one in a homogeneous population; the peak then comes at about  $1 - R^{-1/\lambda}$ . In repeated UK contact surveys, a model of persistent and changing activity predicts  $\lambda = 4.6$ – $5.2$  when daily contacts are capped at 50, the survey’s own convention, and  $3.4$ – $3.7$  at a cap of 20, without fitting to epidemic outcomes. Winter 2020–21 US county waves imply  $2.4$ – $5.3$  for infection fatality rates from  $0.5\%$  to  $1\%$ , and about  $3.5$  at  $0.7\%$ .

## 1 Introduction

In 341 US county epidemic waves of winter 2020–21, the median wave reached its peak of incidence when estimated cumulative infections were  $16\%$  of the population. Under a homogeneous model, the growth at the start of those waves implies a median threshold of  $30\%$ . Infections here are estimated from deaths at a fatality rate of  $0.7\%$ , and growth is converted to a reproduction number with a generation interval of mean  $5.5$  days. We call the peak of incidence the *turnover*.

In the spring of 2020 case counts doubled in under a week, which a homogeneous model reads as a reproduction number near  $2.5$  and a threshold of  $60\%$ . Cotton (2020a,b) called the gap between such thresholds and the observed peaks the herd-immunity paradox and argued that it follows from variation in transmission across people, places and time: early growth is dominated by the fastest transmitters, and the turnover by a harmonic mean (Cotton, 2020b).

People with many contacts are infected sooner, and while infectious they generate more exposure, so each infection removes more than its share of future transmission. Britton et al. (2020) and Gomes et al. (2022) made this quantitative for persistent differences in activity or susceptibility. Tkachenko et al. (2021a) added activity that changes over time, named the immunity factor, and showed that the resulting suppression is partly transient; Tkachenko et al. (2021b) modelled each person’s activity as a mean-reverting process.

The effect is summarized by the *immunity factor*  $\lambda$ , the rate at which the effective reproduction number falls relative to the susceptible share. We estimate it in two ways that share no data. The first uses repeated contact surveys and a model of persistent and changing activity, without

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fitting to epidemic outcomes. The second backs a value out of the peaks of US county waves under an assumed fatality rate. The two ranges overlap, and both depend on their assumptions: the contact cap on one side, the fatality rate and the growth proxy on the other. Gomes et al. (2022) obtain an independent value near 3.5 by fitting heterogeneous connectivity to deaths in England and Scotland.

Where contacts were cut by lockdowns or voluntary distancing, peaks came with 1–5% infected. The value of  $\lambda$  implied by such a peak absorbs the drop in contacts and does not measure immunity.

## 2 The turnover rule

*Unequal activity.* The immunity factor and the relations (1)–(2) below are due to Tkachenko et al. (2021a, eqs. 9, 14, 15). Let  $S$  be the susceptible share. In a homogeneous population the effective reproduction number is  $R_e = RS$ . For fixed gamma-distributed activity the power law below is exact, as shown next. In other models we define  $\lambda$  by the initial fall of  $R_e$  and use the power law as an approximation to predict the turnover. When people differ in activity, write

$$R_e = RS^\lambda. \quad (1)$$

Early on,  $R_e \approx R(1 - \lambda(1 - S))$ , so each percentage point of infection lowers  $R_e$  by  $\lambda$  percentage points of  $R$ . Setting  $R_e = 1$  gives the infected share at the turnover,

$$a^* = 1 - R^{-1/\lambda}. \quad (2)$$

The textbook threshold is the case  $\lambda = 1$ . Equivalently, the textbook formula applies with the adjusted reproduction number  $R^{1/\lambda}$ .

Here  $R$  is the initial effective reproduction number of the model. When  $R$  is inferred from growth it also depends on the assumed generation interval, and in a model with changing activity the two need not coincide.

*Gamma activity.* Let activity  $a$  have a gamma distribution with mean one and squared coefficient of variation  $CV^2 = 1/k$ , and let it scale both the chance of infection and the number of people infected. After cumulative force of infection  $Z$ , the susceptible share is  $S = (1 + Z/k)^{-k}$  and  $R_e \propto \mathbb{E}[a^2 e^{-aZ}] \propto (1 + Z/k)^{-k-2}$ . Hence (1) holds exactly with

$$\lambda = 1 + 2CV^2, \quad (3)$$

as in Gomes et al. (2022, eqs. 21–22), who give  $1 + CV^2$  when only susceptibility varies, and Tkachenko et al. (2021a, eq. 14).

*Age structure.* Let  $M_{ij}$  be the mean number of daily contacts of a person in age group  $i$  with people in group  $j$ , and  $n_i$  the population shares. Infected fractions grow along the leading right eigenvector  $v$  of  $M$ ; let  $u$  be the left eigenvector. Depleting susceptibles along  $v$  lowers the leading eigenvalue at the rate

$$\lambda_{\text{age}} = \frac{\sum_i u_i v_i^2 / \sum_i u_i v_i}{\sum_i n_i v_i}. \quad (4)$$

The numerator measures how strongly early infections sit in the groups that drive transmission; the denominator converts to the overall infected share. The same formula applies to any finite type structure, and we use it for age crossed with activity.

*Accuracy of the rule.* Equation (4) is a first-order slope, so (2) is an approximation away from the gamma case. In the joint age and activity models of Section 5, solved numerically to the peak, the rule understates the infected share at the turnover by 0.6 to 1.2 percentage points at  $R = 1.3$ , 1.4 to 2.9 points at  $R = 1.6$ , and 3.6 to 7.0 points at  $R = 2.5$ .

### 3 Activity that changes over time

*A solvable model.* Tkachenko et al. (2021b) let each person's activity follow a mean-reverting square-root process with a correlation time. A two-state version has closed forms. Let each person's activity take two values with long-run shares  $p_1, p_2$ , mean one and variance  $\sigma^2$ , so that  $(a - 1)^2 = c(a - 1) + \sigma^2$  for a constant  $c$  that sets the skew. Activity switches on a Markov clock with generator  $Q$  and relaxes at rate  $\kappa$ . A person with activity  $a$  is infected at rate  $\beta a \Phi$ , where  $\Phi = \sum_k a_k I_k$ , and infectious people recover at rate  $\gamma$ .

**Proposition 1** (Growth, reproduction number and immunity factor). *The early growth rate  $r$  and the basic reproduction number solve*

$$1 = \beta \left( \frac{1}{r + \gamma} + \frac{\sigma^2}{r + \gamma + \kappa} \right), \quad R_0 = \beta \left( \frac{1}{\gamma} + \frac{\sigma^2}{\gamma + \kappa} \right). \quad (5)$$

*During growth, with  $\rho = r/(r + \kappa)$ , the immunity factor is*

$$\lambda = \frac{(1 + \sigma^2 \rho)/\gamma + \sigma^2 (1 + (1 + c)\rho)/(\gamma + \kappa)}{1/\gamma + \sigma^2/(\gamma + \kappa)}. \quad (6)$$

The proof is in Appendix A. The resolvent  $(r + \gamma - Q^\top)^{-1}$  acts on only two directions, the stationary law  $p$  and the deviation  $(a - 1) \circ p$ , which decays at rate  $\kappa$ . People with more contacts are infected sooner and, while infectious, generate more exposure. The second term in (5) is the persistence of an individual's activity over the infectious period: it contributes while a busy phase outlasts the infection.

*Limits.* With  $\kappa = 0$ , (6) is  $\langle a^3 \rangle / \langle a^2 \rangle$ , the value for persistent heterogeneity (Tkachenko et al., 2021a, eqs. 12, 18). As  $\kappa \rightarrow \infty$  it tends to one. For fast switching,

$$\lambda - 1 \approx r K_a, \quad K_a = \frac{\sigma^2}{\kappa}, \quad (7)$$

where  $K_a$  is the Green–Kubo number of activity, the integral of its autocovariance. A fast epidemic outruns the reshuffling of activity and uses up the busy phase before it is refilled.

*Effective variance.* For the prediction in Section 5 we combine a persistent variance  $\sigma_p^2$  with a switching variance  $\sigma_s^2$  of correlation time  $\tau$  through

$$\text{CV}_{\text{eff}}^2 = \sigma_p^2 + \frac{r}{r + 1/\tau} \sigma_s^2, \quad (8)$$

the persistent part acting in full and the switching part in proportion to the share of a busy phase that survives one growth time. This interpolates between the limits of Proposition 1 and is an approximation.

## 4 Mixtures of places

Cotton (2020b) reaches the same conclusion from a population made of many places. To isolate the effect of heterogeneous local rates, consider independent places with equally seeded epidemics, each with transmission rate  $\beta = \bar{\beta} \eta$  where  $\eta$  has mean one across places. Early growth of the total reads a transmission rate  $\beta_g = t^{-1} \log \mathbb{E} e^{t\bar{\beta}\eta} = G \bar{\beta}$  with  $G > 1$ . The susceptible share at the turnover, averaged over places, is (Cotton, 2020b, §§3.1–3.2)  $\mathbb{E}[\gamma/\beta] = J/\bar{R}$  with  $J = \mathbb{E}[1/\eta] \geq 1$ . Together,

$$s^* \approx \frac{JG}{R_g}, \quad (9)$$

larger than the textbook  $1/R_g$  by two convexity adjustments. In Cotton (2020b) the rate follows an Ornstein–Uhlenbeck process, its growth convexity is Vasicek’s bond-price formula, and the long-run growth excess is the variance times the correlation time (Cotton, 2020b, eq. 16). The same holds for a two-state switch. When each place’s rate switches between  $\bar{\beta} \pm \delta$  at rate  $\nu$ , the total grows at the top eigenvalue  $\bar{\beta} - \gamma - \nu + \sqrt{\nu^2 + \delta^2}$ , which is  $\bar{\beta} - \gamma + \delta^2/(2\nu)$  to first order: the growth convexity is again a Green–Kubo number.

## 5 The immunity factor from contact surveys

*One-day diaries.* The POLYMOD diaries (Mossong et al., 2008) record the contacts of 7,290 people in eight European countries on one day. Diary space limited entries to 29 contacts (30 in the United Kingdom), and in four countries respondents estimated their professional contacts rather than listing them. We count listed contacts, which understates the upper tail. Within age groups the squared coefficient of variation of the number of contacts is 0.53 for ages 0–19, 0.63 for 20–64 and 0.77 for 65 and over. Combining these with the age matrices of Prem et al. (2021) through (4), with activity held fixed, gives  $\lambda = 2.70$  for the United States, 2.77 for Sweden, 2.66 for Spain, 3.02 for the United Kingdom and 2.30 for Brazil. Age alone gives 1.3–1.7.

A one-day diary cannot tell a busy person from a busy day, so these values mix lasting and passing variation. Gomes et al. (2022) estimate the coefficient of variation from POLYMOD and nineteen other surveys, make the same point, and use two-day diaries to separate the two.

*Repeated surveys.* The CoMix survey (Jarvis et al., 2020; Gimma et al., 2022, 2024) ran in the United Kingdom from March 2020 to March 2022. Each participant reported their contacts up to eight or ten times, two weeks apart. We use 21,158 adults aged 18–69 seen at least three times, 160,869 survey days in all. The median participant is observed over 123 days and 90% within 267 days; about 4% rejoin after a gap of around seven months.

Activity is the number of contacts on the diary day divided by the mean for the same week and age group. This removes the common effect of lockdowns and seasons on each age group, though not effects that differ between people. The covariance of a person’s activity between surveys  $\ell$  days apart is fitted as

$$C(\ell) = \sigma_p^2 + \sigma_s^2 e^{-\ell/\tau} \quad (\ell \geq 7), \quad C(0) = \sigma_p^2 + \sigma_s^2 + \sigma_n^2, \quad (10)$$

with a persistent part, a switching part of correlation time  $\tau$ , and a residual  $\sigma_n^2$  that repeat observations at lags of a week or more cannot resolve.

Under this covariance model, 26–30% of the variance of capped daily contacts is persistent and 13–14% belongs to a component with a correlation time of 23 to 36 days. The remaining

| contacts capped at | one-day $CV^2$ | persistent $\sigma_p^2$ | switching $\sigma_s^2$ | $\tau$ (days) | residual $\sigma_n^2$ |
|--------------------|----------------|-------------------------|------------------------|---------------|-----------------------|
| 20                 | 1.82           | 0.54 (30%)              | 0.23 (13%)             | 36            | 1.05 (57%)            |
| 50                 | 3.90           | 1.06 (27%)              | 0.51 (13%)             | 28            | 2.34 (60%)            |
| 100                | 6.54           | 1.69 (26%)              | 0.88 (14%)             | 23            | 3.97 (61%)            |

Table 1: Decomposition of adult activity in the CoMix UK panel.

57–61% is unresolved by repeat observations at lags of a week or more. It includes day-to-day variation, and it may also include genuine correlations lasting less than a week, which could matter over an infectious period. The shares are similar under the three caps, but the levels are not. The fitted covariance is flat from one month to the longest lags, so the persistent component behaves like a habit over the months each person is observed. Restricting the fit to lags of at most 200 days changes each share by at most one percentage point.

*Prediction.* Inserting (8) into the joint age and activity structure with the UK matrix gives the immunity factors in Table 2, for growth rates from 0.02 to 0.2 per day. The residual is left out, so these values treat all sub-weekly variation as irrelevant to depletion. No epidemic outcome enters.

| contacts capped at | $\lambda$ | turnover at $R = 1.6$ | turnover at $R = 2.5$ |
|--------------------|-----------|-----------------------|-----------------------|
| 20                 | 3.4–3.7   | 12–13%                | 22–24%                |
| 50                 | 4.6–5.2   | 9–10%                 | 16–18%                |
| 100                | 5.9–6.8   | 7–8%                  | 13–14%                |

Table 2: Immunity factor predicted from the CoMix decomposition and the UK age matrix, without fitting to epidemic outcomes.

The immunity factor changes by only about 7% across a tenfold range of growth rates, because the switching part is small. The level depends on how the few very high contact counts are treated: it rises from 3.4–3.7 at a cap of 20 to 5.9–6.8 at 100. The CoMix team caps at 50 contacts a day, the value they found to track changes in the reproduction number most closely (Gimma et al., 2022). How many of a very busy person’s contacts can transmit is unresolved.

## 6 The immunity factor in epidemics

*US county waves.* We identify 471 epidemic waves in 372 US counties of more than 50,000 people from daily deaths (New York Times, 2023), February 2020 to June 2021. The data count deaths by the date they were reported, which adds noise to the timing of death peaks. For each wave, early growth  $r$  gives  $R$  through the relation  $R = 1/M(-r)$  of Wallinga and Lipsitch (2007), with  $M$  the moment generating function of an assumed gamma generation interval of mean 5.5 days and standard deviation 2.1 days. Cumulative deaths divided by an infection fatality rate give the infected share when the wave began,  $a_0$ , and at the death peak,  $a_1$ .

Suppose contacts stayed at their early-wave level. Then the wave’s growth and its infected

share at the peak imply, through (2), the value

$$\hat{\lambda} = \frac{\ln R}{-\ln((1 - a_1)/(1 - a_0))}, \quad (11)$$

which we call the *implied peak index*. It equals the immunity factor only if transmission per contact and the number of contacts did not change during the wave. When contacts fell,  $\hat{\lambda}$  absorbs the fall and does not measure immune selection.

Between 95% and 100% of waves in every season turned over below the homogeneous threshold. In winter 2020–21, with 341 waves and growth estimated from deaths, a constant  $\hat{\lambda}$  fits at 3.5 for a fatality rate of 0.7%, 2.4 at 0.5% and 5.3 at 1%. With growth estimated from cases it is about 2.1 at 0.7%.

*The slope is not identified.* Grouped by growth rate, the winter waves infected nearly the same share of their susceptibles whatever their  $R$ . Growth read from deaths and from cases three weeks earlier correlate at only 0.23 in winter, and noise in a regressor flattens any fitted slope. With case growth as an instrument the slope of log depletion on  $\log \ln R$  is 0.91 in spring (90% interval 0.46 to 1.50) and 2.9 in winter (0.7 to 12.5). The county data identify the level of  $\hat{\lambda}$  but not its dependence on growth, which Table 2 says is small.

*Serology.* Antibody surveys measure the infected share directly. Scaling each survey back to the death peak with cumulative deaths gives the infected share at the turnover and the implied peak index in Table 3.

| place              | survey                       | contacts  | $R$     | infected          | $\hat{\lambda}$ |
|--------------------|------------------------------|-----------|---------|-------------------|-----------------|
| England, 9 regions | Ward et al. (2021)           | lockdown  | 2.1–3.5 | 0.9–5.0%          | 17–94           |
| Geneva             | Stringhini et al. (2020)     | closures  | 3.3     | 5.1%              | 23              |
| Stockholm          | Castro Dopico et al. (2021)  | voluntary | 2.8     | 2.3–3.1%          | 33–44           |
| Spain, spring      | Pollán et al. (2020)         | lockdown  | 1.9–4.3 | 0.3–4.6%          | about 60        |
| Spain, autumn      | Ministerio de Sanidad (2020) | curfew    | 1.1–1.5 | 2.6% <sup>†</sup> | about 7         |
| Manaus             | Buss et al. (2021)           | limited   | 1.6     | about 17%         | 2.5             |

Table 3: Turnovers anchored by serology; the infected share is at the turnover. <sup>†</sup>Share of those still susceptible after the first wave.

Where contacts were cut, by law or voluntarily, the first wave turned over with 1–5% infected and  $\hat{\lambda}$  exceeds 15. These values are consistent with the observed reduction in contacts and cannot be read as immunity. Where control was weaker, in Manaus and in the US county waves of winter 2020–21,  $\hat{\lambda}$  is 2 to 3.5. Tkachenko et al. (2021a) fit a transient immunity factor of about 4 to New York City and Chicago during stay-at-home orders, with a model that includes mitigation. Buss et al. (2021) report that distancing in Manaus worked together with immunity.

## 7 Discussion

Contact surveys predict an immunity factor of 4.6–5.2 at the survey’s own cap of 50 contacts a day, 3.4–3.7 at a cap of 20 and 5.9–6.8 at 100. Winter county waves imply a peak index of 2.4–5.3 over fatality rates from 0.5% to 1%, 3.5 at 0.7%, and about 2.1 with growth from cases.

Gomes et al. (2022) fit about 3.5 to England and Scotland. The ranges overlap. At  $R = 2.5$ ,  $\lambda = 3.5$  puts the turnover near 23% and  $\lambda = 5$  near 17%.

On the contact side the level depends on how very high contact counts are treated and on the residual below one week. On the epidemic side it scales with the assumed fatality rate and the growth proxy. The two sets of uncertainties are of similar size.

In the CoMix data most of the resolved heterogeneity persists over the months each person is observed. The switching part is smaller and decorrelates over about a month, so in this model a first-wave turnover is mostly a lasting reduction of the threshold with a transient on top.

Tkachenko et al. (2021a) reach a different balance. They put the long-run immunity factor near 2 and conclude that most of the first-wave suppression was transient. A survey that follows the same people for years would settle how long the persistent component lasts.

The reported later attack in Manaus, 66–76%, is difficult to reconcile with a fixed-activity model of this size. Changing activity is one possible explanation. The contact distribution in Manaus and the blood-donor estimates are additional uncertainties.

The prediction can be tested out of sample. CoMix surveys ran in 21 European countries (Wong et al., 2023), though adult participants in most of them reported only seven times. Fixing the treatment of high contact counts in advance, each survey predicts an immunity factor that the country’s serology and death curves can confirm or refute.

## A Proof of Proposition 1

Let  $I = (I_1, I_2)$  be the infectious people by current activity and  $\Phi = a^\top I$ . Early on,  $\dot{I} = \beta \Phi (a \circ p) - \gamma I + Q^\top I$ . Seek  $I = v e^{rt}$ :

$$v = \beta \Phi (r + \gamma - Q^\top)^{-1} (a \circ p). \quad (12)$$

Since  $Q^\top p = 0$  and  $Q^\top ((a-1) \circ p) = -\kappa (a-1) \circ p$  for a two-state chain, and  $a \circ p = p + (a-1) \circ p$ ,

$$(r + \gamma - Q^\top)^{-1} (a \circ p) = \frac{p}{r + \gamma} + \frac{(a-1) \circ p}{r + \gamma + \kappa}. \quad (13)$$

Taking  $a^\top$  of both sides, with  $a^\top p = 1$  and  $a^\top ((a-1) \circ p) = \sigma^2$ , gives (5);  $R_0$  is the case  $r = 0$ .

For the immunity factor, the susceptible deficit  $s = p - S$  solves  $\dot{s} = \beta \Phi (a \circ S) + Q^\top s$ , and during growth  $s = \beta \Phi (p/r + (a-1) \circ p/(r + \kappa))$  to first order, with overall attack  $x = \beta \Phi / r$ . Then  $a \circ s$  has components  $x(1 + \sigma^2 \rho)$  along  $p$  and  $x(1 + (1 + c)\rho)$  along  $(a-1) \circ p$ , using  $(a-1)^2 = c(a-1) + \sigma^2$ . The next-generation number with susceptibles  $S$  is  $\beta a^\top (\gamma - Q^\top)^{-1} (a \circ S)$ , and subtracting its value at  $S = p$  gives  $R_e = R_0 (1 - \lambda x)$  with  $\lambda$  as in (6). The formulas agree with numerical solution of the model: the growth rate to four digits and  $\lambda$  to 0.1–0.3% (04\_activity\_switching.py).

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